

# 3D Optical AOI Metrology & Coplanarity System

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-AOI-001

**NOTICE: SYNTHETIC PROTOTYPE TECHNICAL MANUAL**

This document is artificially generated for the VectorAI demonstration and software development.

The specifications, thresholds, service-life values, maintenance intervals, and operating parameters are synthetic and must not be used for real industrial equipment operation or maintenance.

## 1. Machine Overview & Manufacturing Context

|                   |                                               |                        |                                  |
|-------------------|-----------------------------------------------|------------------------|----------------------------------|
| Equipment Model:  | 3D Optical AOI Metrology & Coplanarity System | Document ID:           | VAI-MAN-AOI-001                  |
| Machine Type Key: | aoi-inspection                                | Cleanroom Bay / Stage: | Bay 5A: 3D Optical AOI Metrology |
| Document Version: | 1.0                                           | Release Date:          | 2026-08-27                       |

High-speed automated optical inspection and 3D laser profilometry line for inspecting molded IC package surface defects, lead coplanarity, ball grid height, and package dimensional tolerances.

**Process Flow:** Molded leadframe strips or BGA trays travel along a dual-lane SMEMA conveyor. Multi-wavelength RGB+W dome illumination and dual-telecentric cameras capture 100MP images while 3D laser triangulation sensors reconstruct micron-level height maps of leads and solder balls.

## 2. Major Subsystems & Critical Components

| Subsystem / Component       | Primary Function                                                  | Key Parameters                                                             | Degradation Indicators                                     |
|-----------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------|
| 3D Laser Triangulation Head | Scans package surface height and lead coplanarity at 50 kHz.      | Height measurement resolution (0.1µm), laser power (mW), sensor temp (°C). | Laser diode power decay, height measurement noise.         |
| Camera Motion Gantry        | Translates optical sensor across package matrix at high speed.    | Gantry acceleration (G), vibration (mm/s), position jitter (µm).           | Gantry vibration rise (>0.2 mm/s), position encoder error. |
| RGBW LED Dome Illuminator   | Provides structured multi-angle illumination for defect contrast. | Illumination uniformity (%), LED pulse timing (µs).                        | Brightness decay (>10%), strobe timing jitter.             |

### 3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

| Sensor Name & ID                                   | Unit | Normal Range | Warning Range | Critical Range | Direction       |
|----------------------------------------------------|------|--------------|---------------|----------------|-----------------|
| Camera Gantry Vibration<br>vibration_camera_gantry | mm/s | 0.02 – 0.15  | 0.15 – 0.35   | 0.35 – 1       | Higher is Worse |
| Illumination Uniformity<br>optical_intensity       | %    | 95 – 100     | 85 – 95       | 50 – 85        | Lower is Worse  |

### 4. Deterministic RUL Model (Weighted Degradation)

**Base Useful Life:** 3,200 operating hours. **Model:**  $RUL = BaseLife * (1 - (0.50 * wear\_gantry\_vib + 0.30 * wear\_opt\_temp + 0.20 * wear\_light))$

| Parameter        | Sensor ID               | Unit | Weight      | Healthy Limit | Critical Limit | Direction       |
|------------------|-------------------------|------|-------------|---------------|----------------|-----------------|
| Gantry Vibration | vibration_camera_gantry | mm/s | 0.50        | 0.1           | 0.4            | HIGHER IS WORSE |
| Optics Temp      | temp_optics             | °C   | 0.30        | 23            | 35             | HIGHER IS WORSE |
| Light Intensity  | optical_intensity       | %    | 0.20        | 98            | 80             | LOWER IS WORSE  |
| TOTAL WEIGHT SUM | -                       | -    | 1.00 (100%) | -             | -              | VERIFIED OK     |

5. Troubleshooting & Anomaly Symptoms Matrix

| Symptom & ID                                              | Severity | Related Sensors                            | Possible Root Causes                                                                                                              | Recommended Corrective Action                                      |
|-----------------------------------------------------------|----------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| False Defect<br>Classification Rate > 1.5%<br>SYM-AOI-001 | MEDIUM   | vibration_camera_gantry, optical_intensity | <ul style="list-style-type: none"><li>• Gantry rail vibration</li><li>• LED strobe brightness decay</li><li>• Lens dust</li></ul> | Clean telecentric lens and execute standard 3D target calibration. |

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

|                                                                       |                                                                              |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------|
| Scenario 1: Camera Gantry Motion Jitter & Optical Blur (SCEN-AOI-001) | Severity: WARNING                                                            |
| Telemetry Sensor Pattern:                                             | vibration_camera_gantry > 0.25 mm/s                                          |
| Possible Root Causes:                                                 | Linear motor bearing wear   Air-bearing pressure drop                        |
| Recommended Corrective Action:                                        | Inspect air-bearing CDA pressure and calibrate linear motion encoder.        |
| Verification & Recovery Steps:                                        | Verify gantry repeatability (< 0.5µm) > Inspect test target resolution chart |